

- It is impervious to adverse environmental conditions of dust, grease, moisture, pressure, vibrations, chemical or oil vapours.

However, none of the touchscreens currently available have all of above-mentioned characteristics. Accordingly, selection of any touchscreen panel technology is a compromise that will depend on the particular application and environment. Nowadays, the touchscreen interface is progressively becoming a standard requirement in today's electronic products.

The technology within the touch panels

There are a variety of touchscreen technologies with different methods of sensing touch: resistive, capacitive, projected capacitive, scanning infrared (IR) and surface acoustic wave (SAW). Selection of the proper touch technology for a specific application requires evaluation of each type. All touchscreens are composite units comprising a touchscreen overlay and an information-processing interface controller.

Resistive technology. Five-wire resistive touchscreen is the most widely-used touch technology today. A resistive touchscreen panel is basically composed of a glass panel covered with thin conductive and resistive metallic layers, separated by a thin space. So when you touch the screen, two layers touch at that point. Change in the electrical field is detected by the computer and touch point is calculated. Resistive touchscreens are generally the most affordable, but offer about 70 per cent to 85 per cent image clarity.

The screen can be activated with virtually anything rigid (finger, stylus, gloved hand, pen, etc), and the touch has tactile feel. The screen is resistant to surface contaminants and liquids (dust, oil, grease, moisture).

Major benefits of resistive touch technology are its low cost and low

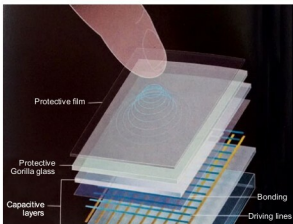


Fig. 1: Touchscreen technology

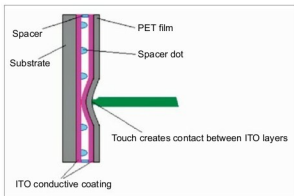


Fig. 2: Contact being made to a resistive touch panel

power consumption. There are three types of resistive touch panels available, namely, four-wire, five-wire and eight-wire.

Surface-capacitive technology.

A surface-capacitive touch panel works by measuring current. Here, a protective cover is placed over the transparent electrode film on top of the glass substrate. Voltage is ap-

plied to all corners, and any contact with the surface disrupts the current. Hence, the touch point is determined by the controller. It is mostly used in the gaming industry. Also, light transmission averages at 80 per cent or above.

This technology offers better image clarity, excellent high resistance to scratching of screen and surface contaminants and liquids (dust, oil, grease, water droplets). A capacitive touchscreen can be operated by the touch of human skin or a stylus holding an electrical charge.

The screens are susceptible to EMI/RFI from the display and environment; higher level of interference can demand more performance from the control-

ler in identifying and rejecting noise that mimics touchscreen activation. These are moderately durable and have limited resolution.

Projected capacitance technology.

Projected capacitive technology is the new powerhouse in the market. This technology is basically used within iPhones and iPads. In this technology, the electronic field is projected through the top of the glass, so energy creates an electrostatic field. So whenever there is change on the top of the surface, change in electrodes is detected, which further calculates the touch point.

It is capable of dealing with two to ten multi-touch points simultaneously depending on the controller being used. Lighting is excellent with a light transmission of 80 per cent or more. But cost is higher as opposed to other technologies.

Also, when you are wearing

General selection criteria for touchscreen panels

There are a dozen or so touchscreen technologies, but makers have mostly settled on capacitive touchscreens because it is the only type of screen that can collect and transmit information for as many as ten spots at the same time. This multi-touch capability permits intricate interactions with the device's display by using, for example, fingers moving apart or together to zoom in or out of an image.